Inverse Function Learning for Image Deblurring: Classical vs. Modern Machine Learning

Progress Report

Surya Manoj Pentakota (2026701036) Kevin Paul Reddy Janagari (2026900006)
IIIT Hyderabad

November 2, 2026

Abstract

This report covers the work completed so far on a controlled comparison of image-deblurring methods that differ in whether they use the physics of the blur. Of the three planned arms, the data-only deep networks (M1) and the physics-aware classical-ML solvers (M2) are fully trained and evaluated on BSDS500 under one locked protocol: 27 000 test cells per method, plus ablations, a perturbation-stability experiment, a perceptual metric and a texture analysis. The unrolled physics-aware networks (M3) are implemented but not yet trained. We describe the algorithms and implementation, the problems we hit and how we addressed them, the remaining work, and what we expect the final evaluation to show. The main finding so far: on the training kernels the two arms are tied, but under kernel mismatch the physics-aware solvers win by up to 6.4 dB, and the data-only networks lose about half of their gain.

Contents

1 Project overview **2**

2 Work completed **2**
2.1 Status at a glance . . . 2
2.2 Timeline . . . 3

3 Experimental protocol **3**

4 Algorithms and models **3**
4.1 M1: data-only networks . . . 3
4.2 M2: physics-aware classical ML . . . 4
4.3 M3: unrolled physics-aware networks (implemented, not trained) . . . 4

5 Implementation details and techniques **5**

6 Results so far **5**
6.1 Main comparison . . . 5
6.2 H2: texture . . . 6
6.3 H3: stability . . . 7
6.4 Perceptual quality (LPIPS) . . . 7
6.5 Hypothesis scorecard (M1 vs. M2; M3 pending) . . . 8

7 Challenges and how we addressed them **8**

8 Remaining work **9**

9 Plan to the final evaluation **9**

1 Project overview

A sharp image x is observed through blur and noise,

$$y = Ax + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I),$$

where A is convolution with a point-spread function (PSF) k. A is ill-conditioned (its spectrum decays to near zero at high frequencies), so naive inversion amplifies noise. We treat deblurring as *inverse function learning*: find g that minimises $\mathbb{E}\|g(y) - x\|^2$. The central question is whether g should be built around the physics of A or learned purely from data. Published methods differ along exactly this axis but are evaluated under different protocols, so we compare them under one.

Three-arm design.

Arm	Family	Knows k, σ?	Primary method (others)
M1	Modern, data-only	No	NAFNet (U-Net)
M2	Classical ML, physics-aware	Yes	ADMM + EPLL-GMM prior (FISTA-TV, Wiener, R–L)
M3	Unrolled network, physics-aware	Yes	USRNet (ISTA-Net, LISTA)

M3 is the control that separates two confounded axes: *physics-awareness* (does the method use k and σ?) and *prior depth* (fixed TV, shallow learned GMM, deep learned CNN).

Hypotheses.

- **H1:** physics-awareness pays off as information degrades: the gap grows with noise, texture complexity and kernel mismatch.
- **H2:** prior depth matters most on texture: M2 is competitive on piecewise-constant content, and deep priors win on natural texture.
- **H3:** physics-awareness buys stability: M1 is unstable under tiny perturbations, and the physics-aware arms resist them.

2 Work completed

2.1 Status at a glance

Component	Content	Status
Shared foundation	FFT operator, PSF library, degradation, BSDS500 loader, locked protocol, metrics	**done**
M1 training	NAFNet, U-Net, five per-σ specialists	**done**
M1 evaluation	Full test set (27 000 cells each), LPIPS, stability	**done**
M2 priors	EPLL GMMs with $K \in \{1, 4, 16, 64, 200\}$	**done**
M2 tuning	λ per noise level on val (FISTA-TV, ADMM-EPLL)	**done**
M2 evaluation	Full test set for all four solvers, K-sweep, LPIPS, stability	**done**
Analyses	Paired statistics, kernel mismatch, data consistency, σ-oracle, $\sigma = 0$ inverse crime, H2 texture, H3 stability, LPIPS, qualitative figure	**done**
Interim report	`report/m1_m2_comparison.md`	**done**
M3	USRNet, ISTA-Net, LISTA implemented and unit-tested	**blocked** (bugs, see §7)
M3 training and evaluation		**pending**

2.2 Timeline

Date	Work
19 Oct	Proposal, design notes, method inventory
20 Oct	Shared foundation; M2 classical solvers; dataset setup; protocol locked (PSF sets, sampling, checkpoint rule, hyperparameters); M1 networks implemented
20–21 Oct	NAFNet trained (100k iterations) and evaluated on the full test set
21 Oct	M3 unrolled networks and the unified evaluator implemented
27 Oct	U-Net, five per-σ specialists and five GMM priors trained; Wiener and R–L evaluated
28 Oct	ADMM-EPLL bug found and fixed; FISTA-TV and ADMM-EPLL ported to the GPU; λ tuning
29 Oct	Full-test M2 evaluation; $\sigma = 0$ quantization check; K-sweep; interim report
30 Oct	Qualitative figure; H3 stability experiment implemented and run; LPIPS
31 Oct	LPIPS finished; H2 texture analysis; report updated

3 Experimental protocol

Every method is evaluated on exactly the same data.

- **Data.** BSDS500, luminance only (Rec. 601) in $[0, 1]$: 200 train / 100 val / 200 test images. Grayscale keeps all arms comparable, since EPLL is a grayscale prior.
- **Noise grid.** $\sigma \in \{0, 1, 5, 15, 25\}/255$.
- **PSF sets.** P_1 (training and in-distribution test): Gaussian $\sigma_k \in \{0.7, 1.2, 1.6, 2.0\}$ and linear motion of length $\{9, 15\}$ px at $\{0, 45, 90, 135\}^\circ$ (12 kernels). P_2-interp: unseen parameters of the same families, Gaussian $\sigma_k \in \{0.9, 1.4, 1.8\}$ and motion of length $\{11, 13\}$ px at $\{22.5, 67.5, 112.5, 157.5\}^\circ$ (11 kernels). P_2-extrap: an unseen family, defocus disks of radius $\{3, 5, 7, 9\}$ px (4 kernels).
- **Forward model.** Circular convolution computed with FFTs; observations are clipped to $[0, 1]$.
- **Seeded observations.** The noise for each (image, PSF, σ) cell is seeded from the cell's names, so every method sees the identical y and results pair one-to-one: $200 \times 27 \times 5 = 27\,000$ test cells per method.
- **Model selection.** Val only. Learned arms keep the checkpoint with the highest mean PSNR over the 60-cell grid on a fixed, seeded 10-image val subset; per-condition selection is not allowed. The test split is never used for selection.
- **Manifest.** PSF membership, splits, sampling policy, checkpoint rule and hyperparameters are written to `artifacts/protocol/` and recorded in every run's metadata with the git commit.

4 Algorithms and models

4.1 M1: data-only networks

The network maps $y \mapsto \hat{x}$ and never sees k or σ.

- **NAFNet** (Chen et al., ECCV 2022), width 32, 17.1 M parameters: a U-Net of NAFBlocks (SimpleGate, simplified channel attention, LayerNorm, no nonlinear activations). Primary M1 model.
- **U-Net**, 7.8 M parameters: the minimal architecture baseline.
- **Per-σ specialists**: NAFNet fine-tuned for 10k iterations at a single noise level. An ablation that measures how much knowing σ would help M1.

Training. One model on $P_1\times$ all noise levels. Each 128^2 patch draws $k \sim \mathcal{U}(P_1)$ and $\sigma \sim \mathcal{U}(\text{grid})$ independently, with random 90° rotations and flips applied before degradation. AdamW ($\beta = (0.9, 0.999)$, weight decay 10^{-4}), L2 loss, batch 16, learning rate 2×10^{-4} with 1k warm-up and cosine decay, gradient clipping 1.0, 100k iterations, validation every 2k, early-stopping patience 4, bfloat16 autocast. NAFNet reached 29.67 dB val PSNR (98k iterations, 2.8 h on an RTX 3090 Ti); U-Net reached 28.43 dB (94k, 0.5 h).

4.2 M2: physics-aware classical ML

The solvers receive the true k and σ and solve the MAP problem

$$\hat{x} = \arg\min_x \tfrac{1}{2}\|Ax - y\|_2^2 + \lambda R(x).$$

- **Wiener filter.** Closed form in the Fourier domain, $\hat{X} = \overline{K}Y/(|K|^2 + \lambda)$, with λ set from σ. Not tuned.
- **Richardson–Lucy.** Multiplicative EM for Poisson likelihood, $x \leftarrow x \odot A^\top(y/Ax)$; 30 iterations. Not tuned.
- **FISTA-TV.** Accelerated proximal gradient (Beck & Teboulle, 2009) with isotropic total variation $R(x) = \sum_i \|\nabla x_i\|_2$. Step $1/L$ with $L = \max |K|^2$; the TV proximal step is solved by ADMM (30 inner iterations, periodic boundary); 100 outer iterations.
- **ADMM-EPLL** (primary). Half-quadratic splitting with the EPLL patch prior (Zoran & Weiss, 2011), $R(x) = -\sum_p \log p(P_p x)$, where p is a Gaussian mixture over zero-mean 8×8 patches:

$$\begin{aligned} x_t &= \arg\min_x \tfrac{1}{2}\|Ax - y\|^2 + \tfrac{\beta_t}{2}\|x - z_{t-1}\|^2 && \text{(closed form, FFT)} \\ z_t &= \text{EPLL-denoise}(x_t;\ s_t), \qquad \beta_t = \lambda/s_t^2. \end{aligned}$$

  The denoiser picks the most likely mixture component for each patch and applies its Wiener filter $(\Sigma_k + s^2 I)^{-1}(\Sigma_k p + s^2 \mu_k)$, then averages overlapping patches (stride 1). The denoiser strength s_t decreases geometrically from 49/255 to $\max(\sigma, 1/255)$ over 10 iterations (the IRCNN/DPIR schedule).
- **GMM prior.** Full-covariance mixture fitted by EM (k-means++ initialisation, 20 iterations) on 100k zero-mean patches from BSDS500 train. $K = 200$ is the primary prior (7 minutes on CPU); $K \in \{1, 4, 16, 64\}$ form the prior-capacity sweep.
- **Tuning.** For FISTA-TV and ADMM-EPLL, λ is chosen per noise level on the 10 val images over P_1, from 10 multiples (2^{-5} to 2^4) of a default. M2 therefore has five tuned scalars per method.

4.3 M3: unrolled physics-aware networks (implemented, not trained)

- **USRNet** (Zhang et al., CVPR 2020), 4.5 M parameters: 8 unrolled HQS iterations alternating the closed-form FFT data step with a ResU-Net prior; a small MLP maps σ to the per-iteration (μ_t, λ_t). Primary M3 model.
- **ISTA-Net**, 0.19 M parameters: unrolled ISTA with a learned convolutional transform and soft-thresholding per iteration.
- **LISTA**, 117 parameters: unrolled ISTA with a shared 1×1 analysis/synthesis pair; the smallest rung of the prior-depth ladder.

Training follows M1 (one model on $P_1\times$ all noise levels) with batch 8, learning rate 10^{-4} and at most 30k iterations; unlike M1, the network receives the true PSF spectrum and σ.

5 Implementation details and techniques

- **Code layout.** `src/deblur/` holds the shared package: `operators/` (FFT convolution), `psf/`, `data/`, `protocol/`, `metrics/`, `methods/{direct,classical,unrolled}/`, `evaluation/` and `training/`. Scripts in `scripts/` cover dataset setup, training, evaluation and analysis. The unittest suite has 133 tests.
- **Unified evaluator** (`scripts/evaluate.py`). Scores any arm on the same cells and also scores the unrestored observation, so every row carries a paired *gain over blurred*. It records the arm, checkpoint, λ values, protocol manifest and git commit. We added: `--subset` (seeded image subset; `--subset 10` on val reproduces the protocol val subset exactly), `--shard` (split across processes), `--sigmas`, `--quantize-input` and `--m2-backend torch`.
- **GPU backend for M2.** FISTA-TV and ADMM-EPLL were re-implemented in PyTorch operation for operation (`methods/classical/torch_backend.py`). FFTs and TV run in float64. The GMM step uses float32 batched matrix products with per-noise-level Cholesky and Wiener factors computed in float64; patch extraction and overlap averaging use `unfold`/`fold`. Unit tests check both solvers against the NumPy reference.
- **Metrics.** PSNR; SSIM; data-consistency residual RMSE($k*\hat{x}-y$); LPIPS (AlexNet, grayscale replicated to three channels); PGD stability.
- **Statistics.** Paired comparisons use the *image* as the unit (PSNR averaged over a set's PSFs), giving $n = 200$ pairs per (set, σ); Wilcoxon signed-rank tests with Holm correction.
- **Stability (H3).** `metrics/stability.py` and `scripts/stability_test.py`: 20-step PGD on NAFNet within an L2 ball $\|\eta\|_2 \le \epsilon\|y\|_2$; the same $y+\eta$ is then given to ADMM-EPLL (a transfer attack).
- **Texture (H2).** Per-image texture score (mean gradient magnitude of the sharp image), terciles, and Spearman correlation with the paired NAFNet$-$EPLL difference.
- **Qualitative figure.** Each example is the test image with the *median* EPLL$-$NAFNet difference for its cell, cropped to the most detailed window, so nothing is chosen by eye.
- **Reproducibility.** Every table in this report and the interim report is generated by a script (`analyze_m1_m2.py`, `analyze_texture.py`, `analyze_h3_lpips.py`, `make_qualitative.py`, `make_progress_tables.py`) from the per-cell evaluation CSVs.

6 Results so far

All numbers are on the 200-image test split and exclude $\sigma = 0$ unless noted (see §7, inverse crime). The interim report (`report/m1_m2_comparison.md`) has the full set of tables.

6.1 Main comparison

- **In distribution the arms are tied:** NAFNet 27.48 vs. ADMM-EPLL 27.43 dB. NAFNet wins at $\sigma \ge 5/255$ by 0.2–0.5 dB; ADMM-EPLL wins at $\sigma = 1/255$ by 0.9 dB.
- **Under kernel mismatch ADMM-EPLL wins at every noise level**, by up to 6.4 dB, and all differences are significant ($p_{\text{Holm}} < 10^{-4}$). NAFNet's gain over the blurred input halves (+5.38 to +2.73 and +2.49 dB); ADMM-EPLL keeps almost all of its gain.
- **The M1 failure is on motion blur.** At unseen motion angles NAFNet loses 3.6 dB, while on unseen Gaussian widths it loses 0.7 dB.
- **The missing information is the kernel, not the noise level:** per-σ specialists gain at most 0.25 dB.

Table 1: Mean test PSNR (dB). Best restored result per row in bold.

			M2 (knows k, σ)				M1 (data only)	
Set	σ	Blurred	Wiener	R–L	FISTA-TV	ADMM-EPLL	U-Net	NAFNet
P_1	1	24.52	25.38	27.98	31.00	**31.89**	29.56	31.00
	5	23.93	23.37	25.10	27.02	27.82	27.44	**28.04**
	15	21.26	21.09	19.27	24.91	25.51	25.56	**25.92**
	25	18.69	18.38	15.80	23.98	24.49	24.64	**24.95**
P_2-interp	1	24.09	24.74	27.27	30.34	**31.24**	24.92	24.85
	5	23.58	23.53	25.11	26.76	**27.48**	24.72	24.79
	15	21.09	21.16	19.56	24.75	**25.29**	24.35	24.51
	25	18.60	18.41	16.04	23.82	**24.32**	23.94	24.13
P_2-extrap	1	22.57	23.27	24.74	27.67	**28.83**	23.00	22.90
	5	22.22	22.97	23.98	24.86	**25.50**	23.58	23.62
	15	20.26	20.74	21.05	23.48	**23.91**	23.35	23.44
	25	18.08	18.06	18.51	22.79	**23.27**	23.02	23.15

Table 2: Kernel-mismatch summary, mean over $\sigma \in \{1, 5, 15, 25\}/255$. Δ is the change from P_1; "gain" is PSNR minus the blurred input.

		P_1	P_2-interp		P_2-extrap		Gain over blurred		
Method	Arm	PSNR	PSNR	Δ	PSNR	Δ	P_1	P_2-int	P_2-ext
Blurred input	–	22.10	21.84	-0.26	20.78	-1.32	–	–	–
Wiener	M2	22.06	21.96	-0.10	21.26	-0.80	-0.04	+0.12	+0.48
Richardson–Lucy	M2	22.04	22.00	-0.04	22.07	+0.03	-0.06	+0.16	+1.29
FISTA-TV	M2	26.73	26.42	-0.31	24.70	-2.03	+4.63	+4.58	+3.92
ADMM-EPLL	M2	27.43	27.08	-0.35	25.38	-2.05	+5.33	+5.24	+4.60
U-Net	M1	26.80	24.48	-2.32	23.24	-3.56	+4.70	+2.64	+2.45
NAFNet	M1	27.48	24.57	-2.91	23.28	-4.20	+5.38	+2.73	+2.49

- **GMM capacity saturates:** $K = 1 \to 4$ adds 1.8 dB, $K = 64 \to 200$ adds 0.02 dB, and the mismatch gap is the same for every K.

6.2 H2: texture

Table 3: Mean test PSNR (dB, $\sigma > 0$) by texture tercile of the test images.

Set	Texture	FISTA-TV	ADMM-EPLL	U-Net	NAFNet	NAFNet − EPLL
P_1	low	29.40	30.19	29.63	30.34	+0.15
	medium	26.65	27.33	26.71	27.38	+0.05
	high	24.10	24.72	24.02	24.67	-0.04
			Spearman ρ(texture, NAFNet−EPLL) $= -0.54$, $p = 7.7 \times 10^{-17}$			
P_2-interp	low	29.16	29.88	27.38	27.47	-2.41
	medium	26.34	26.99	24.35	24.41	-2.58
	high	23.71	24.33	21.68	21.79	-2.53
			Spearman ρ(texture, NAFNet−EPLL) $= -0.09$, $p = 0.23$			
P_2-extrap	low	27.56	28.24	26.13	26.16	-2.08
	medium	24.57	25.26	23.09	23.13	-2.14
	high	21.93	22.59	20.44	20.50	-2.08
			Spearman ρ(texture, NAFNet−EPLL) $= +0.03$, $p = 0.70$			

Contrary to H2, NAFNet's in-distribution lead comes from smooth images (+0.15 dB) and dis-

appears on textured ones (-0.04 dB; $\rho = -0.54$, $p < 10^{-16}$). Under mismatch, texture plays no role.

6.3 H3: stability

Table 4: PSNR drop (dB) when the NAFNet-optimised perturbation of relative size ϵ is given to each method (transfer attack; 10 test images, every PSF; 3 240 attacked cells). Last column: share of cells where ADMM-EPLL drops less. The perturbations are 0.64, 1.27 and 2.51 (/255) RMS.

		NAFNet, ϵ			ADMM-EPLL, ϵ			
Set	σ	0.5%	1%	2%	0.5%	1%	2%	EPLL less (%)
P_1	1	9.84	13.08	16.31	5.59	10.13	15.03	92
	5	4.11	8.55	12.48	1.34	2.82	5.89	99
	15	1.36	3.46	7.38	0.40	0.90	2.04	100
	25	0.85	2.23	5.19	0.21	0.42	0.98	100
P_2-interp	1	5.37	7.14	9.62	3.34	8.27	14.33	38
	5	2.99	5.63	8.83	0.42	1.27	3.25	99
	15	1.57	3.30	6.21	0.09	0.28	1.03	100
	25	0.94	2.15	4.74	0.10	0.19	0.50	100
P_2-extrap	1	4.83	7.25	9.50	3.35	8.08	13.21	41
	5	3.03	5.27	8.16	0.24	1.30	3.42	100
	15	1.26	2.81	5.58	0.08	0.17	0.72	100
	25	0.78	1.78	4.12	0.10	0.18	0.38	100

At $\sigma \geq 5/255$ ADMM-EPLL loses 2–17 times less PSNR than NAFNet and drops less in 99–100% of cells, supporting H3. At $\sigma = 1/255$ the picture is weaker: on P_1 ADMM-EPLL still drops less in 92% of cells, but on both P_2 sets it drops *more* than NAFNet in about 60% of cells. Here the perturbation is as large as the noise, and the weakly regularised solver amplifies it. After the attack, ADMM-EPLL's PSNR is still higher than NAFNet's in every row.

6.4 Perceptual quality (LPIPS)

Table 5: Mean LPIPS on a seeded 50-image test subset (lower is better; best in bold).

Set	σ	Wiener	R–L	FISTA-TV	ADMM-EPLL	U-Net	NAFNet
P_1	1	0.396	0.266	0.115	0.109	0.117	**0.094**
	5	0.560	0.450	0.249	0.267	0.221	**0.208**
	15	0.601	0.735	0.348	0.399	0.323	**0.315**
	25	0.633	0.873	0.394	0.467	0.370	**0.367**
	mean	0.547	0.581	0.276	0.310	0.258	**0.246**
P_2-interp	1	0.439	0.285	0.122	**0.116**	0.280	0.305
	5	0.555	0.447	**0.252**	0.282	0.330	0.346
	15	0.589	0.696	**0.352**	0.414	0.382	0.396
	25	0.619	0.833	**0.399**	0.480	0.409	0.425
	mean	0.551	0.565	**0.281**	0.323	0.350	0.368
P_2-extrap	1	0.551	0.452	**0.215**	0.215	0.430	0.429
	5	0.571	0.483	**0.342**	0.425	0.480	0.510
	15	0.593	0.585	**0.424**	0.549	0.500	0.544
	25	0.617	0.647	**0.477**	0.600	0.507	0.544
	mean	0.583	0.542	**0.364**	0.447	0.479	0.507

NAFNet is perceptually best on the training kernels, even where PSNR is tied. Under mismatch FISTA-TV is perceptually best in every row except P_2-interp at $\sigma = 1$. PSNR and LPIPS rank the

two M2 solvers differently: ADMM-EPLL wins PSNR by 0.4–1.2 dB, while FISTA-TV wins LPIPS at every $\sigma \geq 5/255$.

6.5 Hypothesis scorecard (M1 vs. M2; M3 pending)

Hypothesis	Prediction	Result so far
H1	Gap grows with kernel mismatch	**Supported**: tied on P_1, +2–6 dB for M2 on P_2
	Gap grows with noise	**Not supported**: M2's advantage shrinks with noise
	Gap grows with texture	**Weakly supported** in distribution only
H2	Deep prior wins on texture	**Not supported**: NAFNet's lead comes from smooth images
H3	Physics-aware arm resists perturbations	**Supported at** $\sigma \geq 5/255$ (transfer attack); weak at $\sigma = 1/255$ on P_1 and reversed on P_2

7 Challenges and how we addressed them

1. **ADMM-EPLL barely deblurred.** The first implementation started the HQS penalty at $\beta_0 = 1$ and doubled it each iteration. Because $|\hat{k}(\omega)|^2 \leq 1$, the data step stayed close to the current prior estimate, which starts at the blurred input: on a noiseless motion-blur cell the solver went from 21.4 to only 21.9 dB, even with $\lambda \to 0$. We diagnosed it by sweeping λ on single cells, then replaced the schedule with the IRCNN/DPIR one (strong denoiser first, small β). The fixed solver reaches 35.2 dB on the same cell, and its tuned λ equals the DPIR default $0.23\sigma^2$ at every $\sigma > 0$. A regression test now requires a noise-free blurred image to improve by more than 5 dB.

2. **M2 was too slow on the CPU.** FISTA-TV took 54–69 s and ADMM-EPLL 30–42 s per image on one core, so the full test set would have taken days. Our first estimates were also too optimistic because the machine has 6 physical cores (12 with hyperthreading): the first tuning jobs took 13–19 h instead of about 4 h. We ported both solvers to PyTorch on the GPU. Against the NumPy reference the maximum pixel difference is 10^{-13} (FISTA-TV) and 2×10^{-7} (ADMM-EPLL), and they run 33× and about 50× faster (2.1 s and 0.8 s per image). The full M2 test evaluation then took about one day instead of several.

3. **The noiseless condition is an inverse crime.** At $\sigma = 0$ the observations are made with exactly the FFT operator that M2 inverts, so the Wiener filter scored 80–100 dB. We verified this by rounding the observations to 8 bits (0.0011 RMS error): Wiener collapses to 5–13 dB and ADMM-EPLL loses 6–9 dB, while the networks are unaffected. $\sigma = 0$ is now excluded from all averages and claims, and reported separately.

4. **λ at the edge of the search grid.** The first grid ($\frac{1}{4}\times$ to $4\times$ the default) left the FISTA-TV optimum outside it at $\sigma = 0$ and $\sigma = 25/255$. We widened it to 2^{-5}–2^{4}; every optimum for $\sigma > 0$ is now inside the grid. ADMM-EPLL at $\sigma = 0$ still prefers the smallest value, an inverse-crime effect.

5. **Shared, contended hardware.** The GPU was shared with an inference server and other jobs, and heavy CPU use by M2 slowed M1 work (fine-tunes went from 15 to 60 minutes; evaluations to 2.5 h). We ran one GPU job at a time at the lowest CPU priority, waited for free GPU memory before each job, launched everything detached from the terminal, and limited each solver process to one BLAS thread.

6. **M3 bugs (open).** USRNet's prior pads inputs to a multiple of 4 but downsamples three times, so it crashes on full-size 481×321 validation images. It also returns the data-step output instead of the prior's, so the last prior pass is unused. ISTA-Net and LISTA have no skip connection, so at initialisation they destroy the input: 4.7 dB at the start and 12.5 dB after 1.5k iterations, well

below the ≈25 dB of the blurred input. M3 was deferred so that the M1–M2 comparison could be finished first.

7. **Test-suite incompatibility (open).** Three M3 unit tests fail on PyTorch 2.14, because `assert_close` now requires `rtol` whenever `atol` is given. The fix is a one-line change per test.

8. **Statistical and presentation pitfalls.** The 27 PSFs of an image are not independent samples, so paired tests use the image as the unit. To avoid cherry-picking, qualitative examples are chosen by the median paired difference, not by eye. Every number quoted in the reports was checked against generated tables; this caught two overstated ranges, which were corrected.

8 Remaining work

#	Task	Est. time	Priority
1	Fix M3: USRNet padding (multiple of 8) and output (return the prior's estimate); add identity-preserving skip connections to ISTA-Net and LISTA; fix the 3 failing tests	0.5 day	core
2	Train USRNet, ISTA-Net and LISTA (30k iterations each)	6–12 h GPU	core
3	Evaluate M3 on the full test set; add M3 to the paired tests, mismatch, texture, LPIPS and qualitative analyses	2–3 h GPU	core
4	H3 with M3: transfer attack from NAFNet, and white-box PGD on M3 (`--attack-arm m3`)	3–4 h GPU	core
5	White-box attack on M2 (differentiable FISTA-TV via the GPU port) for a fair stability comparison	1–2 days	high
6	Protocol fix for $\sigma = 0$: quantized observations or a noise floor, and re-evaluation of affected cells	0.5 day	medium
7	Seed variance: retrain NAFNet with 2 more seeds	6 h GPU	medium
8	Sampling sensitivity: family-balanced kernel sampling for M1 (needs a sampler change)	6 h	low
9	Final report (Markdown and PDF) and presentation	2 days	core
10	Stretch: GoPro real blur (blind), DeepDeblur, K-SVD prior	–	optional

9 Plan to the final evaluation

1. **M3 fixes and training** (days 1–2): items 1–2. Smoke-test each model end-to-end before launching full training, and check that the val curve starts at or above the blurred input.

2. **Three-arm evaluation** (day 3): items 3–4. The full infrastructure exists (unified evaluator, analysis scripts), so M3 slots into every table.

3. **Robustness and sensitivity** (days 4–5): items 5–7, in that order, as time allows.

4. **Write-up** (days 6–7): final report with the three-arm tables, the completed hypothesis scorecard and a discussion of the follow-up method; presentation.

10 Expected outcomes for the final evaluation

- **A complete three-arm comparison under one protocol**: PSNR, SSIM, LPIPS, data consistency, kernel mismatch and stability for M1, M2 and M3 on identical observations, with paired statistics.

- **M3 should combine the strengths of M1 and M2.** USRNet's data step uses the true kernel, so we expect its kernel-mismatch drop to be close to M2's (about −0.35 dB on P_2-interp rather than M1's −2.9 dB). Its deep prior should match or beat NAFNet in distribution. If M3 instead degrades like M1 under mismatch, that would mean the learned prior overrides the physics: an informative negative result.

- **A direct test of H1 and H2.** With M3, physics-awareness (M1 vs. M3) and prior depth (M2 vs. M3) can be separated. The current evidence already points to physics-awareness, not prior depth, as what matters under mismatch.
- **A fuller test of H3.** We expect M3 to be more stable than M1 under a transfer attack, because its data step pins the estimate to the observation, but less stable than M2 under white-box attack. The white-box M2 attack will show whether M2's stability advantage at $\sigma \geq 5/255$ survives a perturbation optimised against M2 itself.
- **A diagnosis that motivates the follow-up method**: a physics-aware network whose prior adapts locally (e.g. GMM-routed experts or spatially varying regularisation), targeting the failure modes the comparison identifies.
- **Reproducible artefacts**: code, locked protocol manifests, per-cell CSVs and the scripts that regenerate every table and figure.